



AI with Rav

Learn AI the way you actually think.



DAY 5 of 30

MACHINE LEARNING

99% accurate and completely useless

WHAT YOU'LL LEARN TODAY

- > The 99% model that finds nothing
- > Why accuracy hides failure
- > Precision vs recall in plain words
- > Which metric to pick



99% accurate and completely useless

In One Line: If 99 of 100 transactions are genuine, a model that says "genuine" every time is 99% accurate — and catches zero fraud.

The model that does nothing

Fraud detection. Out of 1,000 transactions, 10 are fraud.

Build a model that predicts **"not fraud"** every single time. No thinking at all.

Accuracy: **99%**. It caught nothing. It is worthless.

THE MODEL THAT ALWAYS SAYS "NOT FRAUD"



990 genuine · caught correctly by doing nothing

10 fraud · every one missed

ACCURACY

99%

RECALL (fraud caught)

0%

990 right, 10 missed — and every missed one is the only kind that mattered

The two questions that matter

- **Precision** — of the ones I flagged, how many were really fraud? (*Am I crying wolf?*)
- **Recall** — of all the real fraud, how much did I catch? (*Am I missing it?*)

Our do-nothing model has **zero recall**. It caught none of the 10. Accuracy never showed that, because 990 easy correct answers drowned out 10 expensive mistakes.

You must choose which error hurts

Missing fraud costs money. Flagging a good customer costs trust. Cancer screening and spam filters sit at opposite ends of that trade.



Pick the one your business actually feels. Then optimise for it, and say out loud which one you chose.

When classes are imbalanced, accuracy is the metric that makes a useless model look excellent.

